

Order Ephemeroptera

# Mayflies

The most primitive winged insects, mayflies are unique in undergoing a final molt after their wings have formed. Their underwater nymphs live for 2–3 years, but adults do not feed, and often live for just one day. There are about 3,000 species in 25 families.

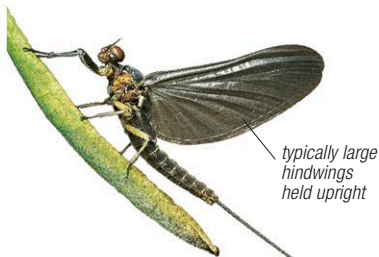
Family Siphonuridae

## Primitive minnow mayflies

**Occurrence** 126 spp. worldwide, but mainly in the Northern Hemisphere; mostly in and near running water



These mayflies have a wingspan of up to 2 in (5 cm). Nymphs are agile swimmers and, when grown, pull themselves out of the water onto a stone or stem.



**SIPHONURUS LACTUSTRIS**  
Anglers call this species the summer mayfly. It is one of the first to colonize a new water course and is tolerant of acidic water.

Family Ephemeridae

## Common burrowing mayflies

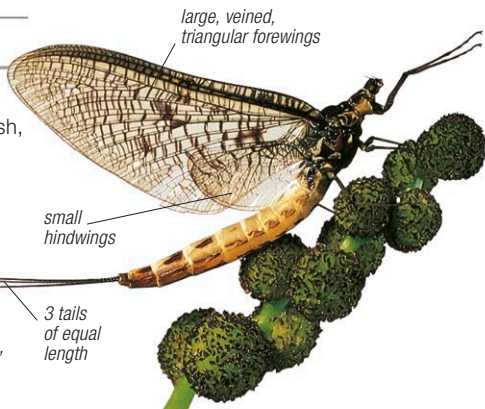
**Occurrence** 96 spp. worldwide except Australia; in and on vegetation around freshwater



These are large mayflies,  $\frac{3}{4}$ –1½ in (1–3.4 cm) long, with clear or brownish, rarely dark-spotted, triangular wings and 2 or 3 long tails at the end of the abdomen. Females lay thousands of

**EPHEMERA DANICA**

Also known as the greendrake, this is a large, widespread, European species. It breeds in rivers and lakes with sandy or silty bottoms.



Order Odonata

# Damselflies and dragonflies

With their large eyes and superb flying ability, these insects are highly effective aerial predators. Damselflies are slender, and tend to rest with wings folded, while the stouter dragonflies rest with wings held out. There are about 5,600 species, in 32 families. All start life as aquatic nymphs.

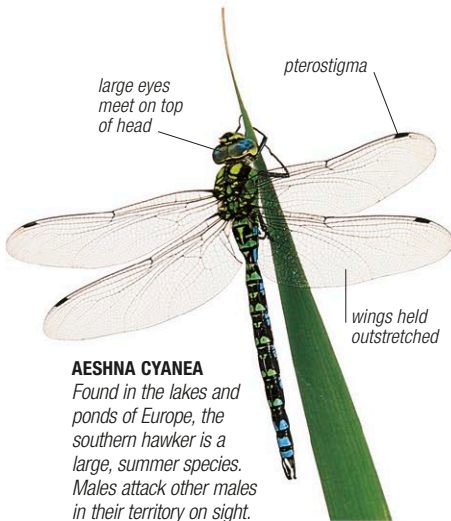
Family Aeshnidae

## Darners

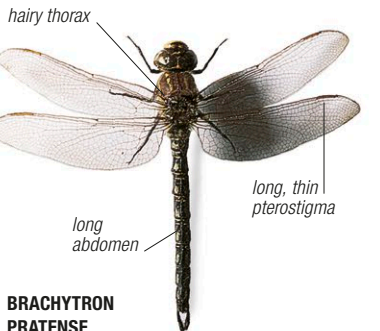
**Occurrence** 444 spp. worldwide; in and near still water, ponds, swamps with vegetation



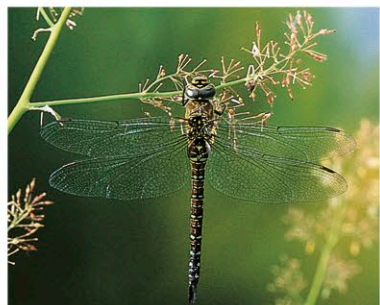
Also known as hawkers, darners are some of the biggest, most powerful dragonflies, with a wingspan of mostly 2½–3½ in (6.5–9 cm). Their large eyes touch on top of the head. Males hold a territory and patrol it regularly, while females cut slits in plants to lay their eggs. Their nymphs are large and well camouflaged, and hunt animals as large as tadpoles and young fish.



**AESHNA CYANEA**  
Found in the lakes and ponds of Europe, the southern hawker is a large, summer species. Males attack other males in their territory on sight.



**BRACHYTRON PRATENSE**  
The adult hairy dragonfly typically emerges in May, flying low, with a characteristic zigzag path.



**AESHNA MIXTA**  
The migrant hawker, widespread in central and southern Europe, follows along a regular beat to catch prey. It flies in summer and fall, and often migrates across Europe in large numbers.

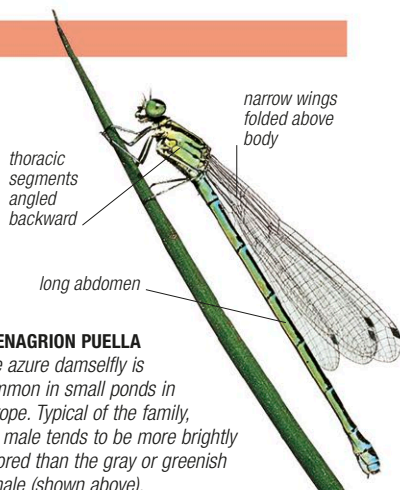
Family Coenagrionidae

## Narrow-winged damselflies

**Occurrence** 1,142 spp. worldwide, especially in temperate regions; in ponds, bogs, streams, brackish water



These slender-bodied damselflies mostly have a wingspan of  $\frac{3}{4}$ –1¼ in (2–4.5 cm). Females make slits in aquatic plants and lay eggs in them. Nymphs hunt on underwater plants.



**COENAGRION PUELLA**

The azure damselfly is common in small ponds in Europe. Typical of the family, the male tends to be more brightly colored than the gray or greenish female (shown above).

Family Lestidae

## Stalk-winged damselflies

**Occurrence** 152 spp. worldwide; in swampy or boggy places, pools, lakes, slow-flowing streams



Also called spread-winged damselflies, these insects have sturdy, bright blue or green bodies, with a metallic sheen. At rest they adopt a vertical pose with the head up and the wings, spanning 1¼–3 in (3–7.5 cm), held out. Males have forceps-shaped claspers at the end of the abdomen. Females lay eggs inside the aerial parts of aquatic plants.



**LESTES SPONSA**

The emerald damselfly is found in Europe and Asia. It has a dark green body and inhabits canals, ponds, and lakes.

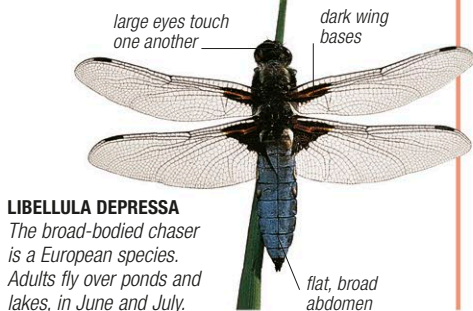
Family Libellulidae

## Common skimmers

**Occurrence** 986 spp. worldwide; in forests near slow-flowing streams, ponds, bogs



Also known as darters due to their unpredictable flight pattern, these stout-bodied, often colorful, insects have a wingspan of mostly 1¾–3 in (4.5–7.5 cm). The broad, flat abdomen makes them highly maneuverable. Females hover over water and dip down to release their eggs. Nymphs hunt for prey in mud and debris.



**LIBELLULA DEPRESSA**

The broad-bodied chaser is a European species. Adults fly over ponds and lakes, in June and July.